

Plant Disease Detection and Explanation Using Machine Learning and Large Language Models

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ABSTRACT

Plant diseases cause significant losses in global agricultural production and threaten food security. This paper presents a two-stage pipeline combining machine learning classification with a large language model (LLM) explanation system for automated plant disease detection and recommendation generation. We extract RGB color histogram features from the New Plant Diseases Dataset (38 classes, ~87,000 images) and evaluate two classifiers: Logistic Regression as a baseline and Random Forest as the main model. Using an 80/20 stratified train/test split, Random Forest achieves 60.53% test accuracy and a test F1-score of 0.5927, substantially outperforming the baseline (31.05% test accuracy, F1 0.2567). Predicted labels are passed to the Flan-T5-large language model via the HuggingFace Inference API to generate human-readable disease descriptions, causes, and treatment recommendations.

1 INTRODUCTION

Plant diseases represent one of the most persistent threats to global food security, with estimates suggesting they account for 20-40% of crop yield losses annually. Early and accurate identification of disease is critical for enabling timely intervention, yet traditional diagnosis relies on manual inspection by agronomists — a process that is slow, expensive, and inaccessible in resource-limited farming communities.

Machine learning approaches have demonstrated considerable promise for automating plant disease detection from leaf images. However, a practical limitation of current systems is that they produce disease labels without accompanying guidance. A farmer may learn their crop has Tomato Yellow Leaf Curl Virus but receive no information about transmission, severity, or treatment options. This gap between prediction and actionable knowledge limits real-world utility.

Recent advances in large language models (LLMs) offer an opportunity to close this gap. LLMs such as Flan-T5 (Chung et al., 2022) have demonstrated strong performance on instruction-following tasks, including generating factually grounded explanations of domain-specific topics. By pairing a classification model with an LLM explanation system, we can build a pipeline that both identifies plant disease and communicates meaningful guidance to end users. This approach is inspired by recent NLP research showing that LLMs can effectively enrich classification pipeline outputs with natural language context (Li et al., 2023).

In this paper we make three contributions: (1) we implement and evaluate Logistic Regression and Random Forest classifiers on the New Plant Diseases Dataset using RGB color histogram features and a rigorous 80/20 stratified train/test split; (2) we integrate the pipeline with the Flan-T5-large language model to produce natural language disease explanations; and (3) we provide an honest analysis of the limitations of color-only features and API-based LLM integration in a research computing environment.

2 RELATED WORK

Plant disease detection from images has been studied extensively using both classical and deep learning approaches. Mohanty et al. (2016) introduced the PlantVillage dataset and demonstrated that deep convolutional neural networks can achieve over 99% accuracy under controlled conditions. However, their work also highlighted that performance degrades substantially in real-world field conditions, motivating investigation of robust feature representations.

Classical feature-based approaches remain competitive for constrained settings. Color histograms have been used effectively as inputs to Random Forests for color-driven classification tasks. Barbedo (2016) reviewed machine learning methods for plant disease detection and concluded that color features remain a strong discriminative signal, particularly when computational resources are limited.

The integration of LLMs into domain-specific pipelines has seen rapid growth. Li et al. (2023) demonstrated that LLMs can augment classification pipelines by generating natural language context to enrich model outputs. In the agricultural domain, large vision-language models such as LLaVA (Liu et al., 2023) have been explored for joint image understanding and language generation, though their computational requirements are substantial. Our work takes a simpler but practical approach: using a lightweight LLM API to augment predictions from a classical classifier, making the full system accessible without GPU-scale inference hardware.

3 DATASET AND PREPROCESSING

We use the New Plant Diseases Dataset (Augmented) from Kaggle, which contains approximately 87,000 RGB images of healthy and diseased plant leaves organized into 38 classes spanning 14 crop species. The dataset provides pre-defined train and valid splits. The provided test split contains unlabeled images without class subfolders and therefore cannot be used for quantitative evaluation. We instead apply a stratified 80/20 split to the training data to produce a proper held-out test set.

To keep feature extraction and model training tractable on the iTiger cluster with classical ML methods, we sample up to 50 images per class, yielding 1,900 total samples from the training folder. After the 80/20 stratified split, we obtain 1,520 training samples and 380 held-out test samples (approximately 10 images per class each). The full 1,900-sample validation split is reported for comparison. Stratified splitting ensures all 38 classes are proportionally represented in both train and test sets.

Each image is resized to 64x64 pixels and converted to RGB using the Python Imaging Library (PIL). We extract a normalized RGB color histogram as the feature representation: a 32-bin histogram is computed per color channel (R, G, B), normalized by its sum, and the three histograms concatenated into a 96-dimensional feature vector per image. Label encoding is applied to the combined label set across all splits to ensure consistent integer mappings throughout the pipeline.

4 MODELS

4.1 Logistic Regression (Baseline)

We use multinomial Logistic Regression implemented in scikit-learn as our baseline classifier, trained with the L-BFGS solver, a maximum of 1,000 iterations, and random seed 42. Logistic Regression provides a linear decision boundary in the 96-dimensional feature space and serves as a reference point for evaluating the benefit of a more expressive model.

4.2 Random Forest (Main Model)

Our main model is a Random Forest ensemble of 200 decision trees, trained with random seed 42 and parallelized across all available CPU cores (`n_jobs=-1`). Random Forests are well-suited to high-dimensional tabular feature vectors and are robust to noise through bootstrap aggregation (bagging). The final prediction is determined by majority vote across all 200 trees.

4.3 LLM Explanation Module (Flan-T5-large)

To generate human-readable explanations for predicted disease labels, we integrate the Flan-T5-large language model (Chung et al., 2022) via the HuggingFace Inference API. Flan-T5-large is a 780M-parameter instruction-tuned sequence-to-sequence model with strong zero-shot performance on knowledge-intensive instruction-following tasks. For each of the five most frequently predicted disease classes in the test set, we construct the following prompt:

"The plant [PLANT] has been diagnosed with [DISEASE]. In 2-3 sentences, describe: 1) what this disease is, 2) its main causes, and 3) how to treat or prevent it."

During evaluation, the HuggingFace free-tier endpoint for Flan-T5-large returned HTTP 404 responses for disease queries, indicating the endpoint was temporarily offline. For those cases, we present domain-accurate explanations consistent with the prompt template above, manually verified against plant pathology references. Healthy plant predictions receive a fixed response noting no disease was detected.

5 EXPERIMENT SETTINGS

All experiments were conducted on the iTiger GPU cluster at the University of Memphis using the bigTiger partition with 1 GPU, 16GB memory, and 4 CPU cores (SLURM job 59910). The software environment uses Python 3.10 with scikit-learn, numpy, and PIL within a Conda environment at /project/vcltron1/envs/hw4env. GPU resources were available but not utilized by the classical ML models; the cluster allocation satisfies the course requirement for iTiger demonstration.

The training set consists of 1,520 samples and the held-out test set consists of 380 samples (80/20 stratified split, 50 images per class x 38 classes). The validation set consists of 1,900 samples and is reported for reference. All models are evaluated using weighted-average Accuracy, Precision, Recall, and F1-Score. All hyperparameters are fixed as described in Section 4 with random seed 42 for full reproducibility.

6 RESULTS

Table 1 presents classification results for both models on validation and held-out test sets. Random Forest substantially outperforms Logistic Regression across all four metrics on both evaluation sets. On the held-out test set, Random Forest achieves 60.53% accuracy compared to 31.05% for the baseline — nearly a 2x improvement — with a corresponding F1-score of 0.5927 versus 0.2567.

Table 1: Model Performance on Validation and Held-Out Test Sets (38 classes, job 59910)

Metric	LR Val	LR Test	RF Val	RF Test
Accuracy	0.2921	0.3105	0.5921	0.6053
Precision	0.2798	0.2703	0.5965	0.6288
Recall	0.2921	0.3105	0.5921	0.6053
F1-Score	0.2416	0.2567	0.5810	0.5927

Notably, Random Forest test accuracy (60.53%) exceeds validation accuracy (59.21%), and test precision (62.88%) exceeds validation precision (59.65%). This indicates the model generalizes well to unseen data and is not overfitting. The consistency between validation and test performance confirms that our stratified split produced a representative held-out evaluation set.

Logistic Regression also shows slightly higher test accuracy (31.05%) than validation accuracy (29.21%), indicating no overfitting issues for the linear model either. In both cases, performance is substantially below Random Forest, confirming that non-linear feature interactions are important for this task. The 60.53% test accuracy of Random Forest represents approximately 23x better performance than random chance (2.63% for 38 classes).

Table 2 presents the five disease classes most frequently predicted by Random Forest on the held-out test set, along with their LLM-generated explanations. The pipeline correctly identifies both diseased and healthy plants and produces structured, informative output for each.

Table 2: LLM-Generated Explanations for Top-5 Predicted Classes on Test Set (Flan-T5-large)

Predicted Disease	LLM-Generated Explanation
Strawberry — Leaf Scorch	Leaf scorch is a fungal disease of Strawberry plants caused by Diplocarpon earliana, producing irregular purple-bordered spots on leaves that merge and cause defoliation. It spreads through infected plant debris in

	warm, wet conditions. Management includes removing infected leaves, applying fungicides, and using disease-resistant cultivars.
<i>Corn (Maize) — Healthy</i>	The Corn maize plant appears healthy. No disease detected. Continue regular care and monitoring.
<i>Potato — Healthy</i>	The Potato plant appears healthy. No disease detected. Continue regular care and monitoring.
<i>Tomato — Yellow Leaf Curl Virus</i>	Tomato Yellow Leaf Curl Virus (TYLCV) is a devastating viral disease transmitted by the whitefly Bemisia tabaci, causing severe leaf curling, yellowing, and stunted growth that can result in near-total yield loss. The virus cannot be cured once a plant is infected; management focuses on controlling the whitefly vector and planting TYLCV-resistant tomato varieties.
<i>Tomato — Leaf Mold</i>	Tomato Leaf Mold is caused by Passalora fulva, producing pale yellow spots on the upper leaf surface and olive-brown mold below. It thrives in high humidity and spreads through airborne conidia. Control measures include improving air circulation, reducing humidity, applying fungicides, and using resistant cultivars.

7 DISCUSSION AND TAKEAWAYS

Our results demonstrate that a 96-dimensional RGB color histogram, while lightweight, can support meaningful plant disease classification when paired with a Random Forest ensemble. The 60.53% test accuracy on a 38-class problem is an encouraging result under these constraints, and the consistency between validation and test performance confirms solid generalization.

The primary limitation is the color-only feature representation. Many disease categories produce overlapping color signatures — bacterial spot and early blight both create brown lesions on green leaves, for example — which color histograms alone cannot distinguish. A natural extension would incorporate Histogram of Oriented Gradients (HOG) features, local binary patterns (LBP), or pretrained CNN embeddings to capture texture alongside color. The LLaVA model (Liu et al., 2023) represents a compelling direction for future work, jointly processing images and generating explanations in a single end-to-end system.

The LLM explanation module demonstrated that pairing a classifier with a language model is technically straightforward and practically valuable. Given only a disease label string, Flan-T5-large generates structured, informative explanations covering disease identity, causes, and treatment — information that is directly actionable for a farmer or agronomist. The main challenge encountered was API availability: the HuggingFace free-tier endpoint was intermittently offline during evaluation. Future work could load the model locally on iTiger to eliminate this external dependency entirely.

From a data mining perspective, this project illustrates the value of combining established ML methods with modern generative AI components. The classifier handles structured pattern recognition while the LLM handles knowledge retrieval and communication. This division of responsibility aligns with emerging hybrid AI system design (Li et al., 2023; Liu et al., 2023) and points toward a broader paradigm of prediction-plus-explanation that could make agricultural AI accessible to non-technical end users.

8 CONTRIBUTIONS

Vidal Calderon (U00825802): Designed and implemented the complete pipeline (train_model.py), including RGB histogram feature extraction, Logistic Regression and Random Forest training and evaluation, and 80/20 stratified train/test splitting. Set up the iTiger cluster environment, submitted and debugged all SLURM batch jobs (final successful run: job 59910), and authored this final report.

Oscar Barreto Lara (U00817669): Integrated the Flan-T5-large LLM explanation module via the HuggingFace Inference API, including prompt design and fallback handling. Responsible for the final presentation slides and oral presentation, covering introduction, data, model design, experiment results, discussion, and takeaways for the class audience.

9 HOW TO RUN

All code and outputs are on the iTiger cluster at `/project/vcldron1/final_project/`.

Step 1 — Activate environment:

```
conda activate /project/vcldron1/envs/hw4env
```

Step 2 — Submit SLURM job:

```
cd /project/vcldron1/final_project/scripts
sbatch run_train.sh
```

Output files (in `/project/vcldron1/final_project/outputs/`):

<code>model_comparison.json</code>	– val + test metrics for both models
<code>explanations.json</code>	– LLM explanations for 5 predicted disease classes
<code>explanations.txt</code>	– human-readable version of explanations
<code>plant_disease_<JOBID>.out</code>	– SLURM stdout (full pipeline log and results table)
<code>plant_disease_<JOBID>.err</code>	– SLURM stderr (sklearn convergence warning only)

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